Haoxiang Wan

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Education

University of Houston, Houston, TX *Aug. 2025 – May 2029 (Expected)*
Ph.D. in Electrical and Computer Engineering
Advisor: Prof. Xingpeng Li, Renewable Power Grid (RPG) Laboratory
Research Focus: Power System Operations and Optimization, Data Center Load Flexibility

University of Illinois at Urbana-Champaign, Urbana, IL *May 2025*
Master of Engineering in Energy Systems

The University of Sheffield, Sheffield, UK *June 2024*
Bachelor of Engineering in Electrical and Electronic Engineering, *First Class Honours*

Publications

- **H. Wan** and X. Li, "Data Center Spatio-Temporal Load Flexibility in Security-Constrained Unit Commitment for Enhanced Grid Efficiency and Reliability," accepted by *IEEE Industry Applications Society (IAS) Annual Meeting*, 2026.
- **H. Wan**, L. Fang, and X. Li, "Grid Operational Benefit Analysis of Data Center Spatial Flexibility: Congestion Relief, Renewable Energy Curtailment Reduction, and Cost Saving," accepted by *IEEE PES General Meeting (PESGM)*, 2026. [arXiv:2511.08759]

Research Experience

Graduate Research Assistant *Aug. 2025 – Present*
Renewable Power Grid (RPG) Laboratory, University of Houston, Houston, TX
Advisor: Prof. Xingpeng Li

- Developed Security-Constrained Unit Commitment (SCUC) models that integrate data center spatial and temporal load flexibility, improving grid efficiency and reliability while relieving transmission congestion and reducing renewable energy curtailment.
- Performed contingency-constrained hosting capacity analysis to identify transmission bottlenecks limiting large-load interconnection on standard IEEE test systems.
- Built data generation pipelines and machine learning models for locational marginal price (LMP) and congestion forecasting in collaboration with Dow.
- Implemented all optimization models in Python/Pyomo, AMPL, and Gurobi.

Undergraduate Capstone: Microcontroller-Based Traffic Signal Control System *Sept. 2023 – June 2024*
The University of Sheffield, Sheffield, UK
Supervisor: Dr. Kennedy Offor

- Designed and implemented an embedded traffic management system programmed in C to optimize urban traffic flow through dynamic signal control.
- Developed a custom four-layer PCB integrating sensor arrays and wireless communication modules.

Teaching Experience

Graduate Teaching Assistant *Jan. 2026 – May 2026*
ECE 5380 – Power Electronics & Electric Drives, University of Houston
Instructor: Dr. Hao Huang

- Supported a synchronous online graduate course covering power semiconductor devices, converters and inverters, DC/induction/synchronous motor drives, harmonics, and filtering.
- Graded homework and examinations for students, providing error-focused feedback to improve learners' comprehension.

Graduate Teaching Assistant *Aug. 2025 – Dec. 2025*
ECE 5388 – Renewable Energy Technology, University of Houston
Instructor: Dr. Hao Huang

- Assisted with instruction and assessment for a course covering solar, wind, and grid integration of renewable resources.
- Provided detailed feedback on assignments and examinations to support student learning outcomes.

Professional Experience

Research Aide (Summer Internship) *May 2026 – Aug 2026*
Argonne National Laboratory, Lemont, IL
Supervisor: Dr. Jonghwan Kwon, Energy Systems and Infrastructure Analysis Division

- Modeling large-scale data center load integration on the ERCOT transmission network to evaluate grid reliability and congestion impacts under accelerated AI-driven load growth.
- Developing optimization frameworks to assess hosting capacity and identify transmission bottlenecks for hyperscale data center deployment in Texas.

Engineering Intern *June 2023 – Aug. 2023*
Jiangxi Engineering Research Center of High-Power Electronics and Grid Smart Metering, China

- Contributed to the design of a specialized current detection device for power grid leakage identification, with MATLAB-based simulations validating its reliability under varying operating conditions.

Engineering Intern *Aug. 2021*
Jiangxi Branch of China Comservice Supply Chain Management Co., Ltd., China

- Performed maintenance, diagnostics, and troubleshooting of Hikvision surveillance equipment, applying principles of electronic circuitry and signal processing to identify and resolve hardware faults.

Technical Skills

- **Programming Languages:** Python, MATLAB, C, C++
- **Optimization & Modeling:** Pyomo, AMPL, Gurobi
- **Power System Tools:** MATLAB/Simulink, LTspice
- **Machine Learning:** PyTorch, scikit-learn, pandas, NumPy
- **Other:** LaTeX, Linux/HPC environments
- **Languages:** English (fluent), Mandarin Chinese (native)

Honors and Awards

- Engineering Excellence Scholarship, The University of Sheffield *2021–2022, 2022–2023*
- NCUK Sheffield Scholarship, The University of Sheffield *2021–2022, 2022–2023*

Professional Affiliations

- Student Member, IEEE Power & Energy Society (PES)
- Student Member, IEEE Industry Applications Society (IAS)